Lecture 7: Data Gathering and Sensemaking

CSE 4849: Human-Computer Interaction

Preece, Rogers & Sharp • Interaction Design (6th ed.) • Chapters 8–11

Learning Objectives

- Plan and execute rigorous data gathering: interviews, questionnaires, and observation
- Apply appropriate analysis techniques to qualitative and quantitative data
- Understand the challenges and ethics of data collection at scale (big data)
- Translate gathered data into actionable design requirements using personas and scenarios
- Evaluate trade-offs between data gathering methods for HCI contexts

gathering → analysis → scale → requirements

The Data Gathering & Sensemaking Lifecycle

- PLAN → Define goals, identify participants, choose methods, obtain consent
- GATHER → Execute interviews, questionnaires, observation, ethnography
- ANALYZE → Code qualitative data, compute quantitative metrics, find patterns
- SCALE → Handle large datasets, visualize trends, ensure ethical compliance
- DESIGN → Transform insights into requirements, personas, scenarios, use cases
- Each stage informs the next; iteration is essential in HCI research

PART 1

Planning & Conducting Data Gathering

Five Key Issues in Data Gathering

- 1. Setting Goals — Decide research questions and how data will be analyzed before collection
- 2. Identifying Participants — Determine target population, sample size, and recruitment strategy
- 3. Relationship with Participants — Establish trust; obtain informed consent; ensure professionalism
- 4. Triangulation — Collect multiple data types (e.g., video + notes + artifacts) to strengthen validity
- 5. Pilot Studies — Run a small trial to refine instruments, timing, and logistics before full deployment

Data Recording Methods

Methods

- Handwritten notes — flexible but selective
- Audio recording — captures verbatim speech
- Video recording — preserves verbal + non-verbal behavior
- Photographs — document artifacts, layouts, environments
- Digital logs — automatic interaction tracking

Considerations

- Multiple methods together increase reliability
- Video is rich but time-consuming to analyze
- Always ask permission before recording
- Consider observer effect: cameras change behavior
- Backup strategy: redundant storage is essential

Interview Types: A Spectrum of Structure

- Unstructured — No script; exploratory, rich, but not replicable; good for early discovery
- Semi-structured — Guided by a script with flexibility to probe emergent themes; balanced approach
- Structured — Tightly scripted, like a verbal questionnaire; highly replicable but may lack depth
- Focus Groups — Group interview; generates discussion and consensus but risks groupthink
- Digital Interviews — Skype, Zoom, email, smartphones; convenient, anonymous, but lose some nuance

Crafting Interview Questions

Question Types

- Closed questions — predetermined answers (yes/no, scales); easy to analyze
- Open questions — free-form responses; rich but labor-intensive to analyze
- Use closed for demographics and facts; open for motivations and stories

What to Avoid

- Long, compound sentences — split into two questions
- Jargon or technical language the interviewee may not understand
- Leading questions that assume behavior (e.g., 'Why do you like...?')
- Unconscious biases (gender, age, cultural stereotypes) in phrasing
- Double-barreled questions that ask two things at once

Running an Interview: Five Phases

- Introduction — Introduce yourself; explain goals; reassure on ethics; ask to record; present consent form
- Warm-up — Begin with easy, non-threatening questions to build rapport
- Main Body — Present questions in logical order; probe interesting responses without derailing
- Cool-off — Include a few easy questions at the end to reduce tension
- Closure — Thank the interviewee; clearly signal the end (e.g., switch recorder off); explain next steps

Questionnaire Design Principles

- Question order matters — early questions prime respondents' thinking for later ones
- Different populations may need different versions (localization, literacy levels)
- Provide clear, explicit instructions on how to complete each section
- Balance white space with compactness — too sparse feels long; too dense feels overwhelming
- Avoid very long questionnaires; respect participants' time
- Decide on polarity: all positive, all negative, or mixed phrasing to avoid acquiescence bias

Response Formats

Discrete Formats

- Yes / No checkboxes — simple but limited nuance
- Multiple checkboxes — allow multiple selections
- Radio buttons — force a single choice
- Dropdown menus — save space for long option lists

Scale Formats

- Likert scales — agreement level (1=Strongly Disagree to 5/7=Strongly Agree)
- Semantic differential scales — bipolar adjectives (e.g., Fast–Slow)
- 3, 5, 7, or more points — odd numbers allow neutral; even numbers force commitment
- Open-ended text boxes — capture unexpected insights but harder to quantify

Online Questionnaires: Advantages & Challenges

Advantages

- Easy and quick to distribute globally
- Responses arrive rapidly; real-time monitoring
- No copying or postage costs
- Data automatically enters a database for analysis
- Errors can be corrected without reprinting
- Branching logic can personalize question paths

Challenges

- Sampling is problematic when population size is unknown
- Preventing multiple responses from the same individual is difficult
- Question tampering can occur in emailed questionnaires
- Self-selection bias: only motivated users respond
- Digital literacy requirements exclude some populations

Observation: Field vs. Controlled Environments

- Field Observation — Study users in their natural context (home, office, hospital); high ecological validity
- Controlled Observation — Study users in a lab or staged setting; easier to record but less authentic
- Direct Observation — Researcher watches in person; can ask clarifying questions
- Indirect Observation — Track via diaries, interaction logs, web analytics, or remote sensors
- Think-Aloud Protocol — User verbalizes thoughts during task; reveals reasoning but slows performance

Structuring Observation: The SPACES Framework

- Space — What is the physical space like and how is it laid out?
- Actors — Who are the people involved and what are their relevant details?
- Activities — What are actors doing and why?
- Objects — What physical artifacts are present (furniture, tools, devices)?
- Acts — What are the specific individual actions?
- Events — Is the observation part of a special or routine event?
- Time — What is the sequence of events?
- Goals — What are actors trying to accomplish?
- Feelings — What is the mood of the group and of individuals?

Three easy-to-remember parts:
The person: Who?
The place: Where?
The thing: What?

- Decide on how involved you will be: from passive observer to active participant
- How to gain acceptance
- How to handle sensitive topics, for example, culture, private spaces, and so on
- How to collect the data:
  - What data to collect
  - What equipment to use
  - When to stop observing

Ethnography & Online Ethnography (Netnography)

- Ethnography — Immersive philosophy of studying culture through participant observation and interviews
- Degree of participation varies from passive observer to active participant
- Requires cooperation of participants; informants are invaluable guides
- Data analysis is continuous; questions refine as understanding deepens
- Online Ethnography (Netnography) — Studies virtual communities and digital behavior
- Online interaction differs from face-to-face; virtual worlds have persistence physical worlds lack
- Ethical considerations differ: public posts vs. private messages; anonymity is harder to guarantee

Combining Data Gathering Techniques

- No single method captures the full picture — triangulation is essential
- Diaries + Interviews — Diaries provide temporal data; interviews provide depth and clarification
- Observation + Think-Aloud + Questionnaire — Observation reveals behavior; think-aloud reveals cognition; questionnaire quantifies attitudes
- Documentation Review + Product Evaluation + User Observation + Group Interviews — Comprehensive organizational analysis
- Ethnographic Study + Usability Tests + User Participation — Deep contextual understanding plus validated interaction design
- Choice depends on: study focus, participants, technique nature, resources, and time available

PART 2

Making Sense of Data: Analysis

Qualitative vs. Quantitative Data

Quantitative

- Expressed as numbers — counts, frequencies, durations, ratings
- Enables statistical analysis: means, medians, modes, standard deviations
- Supports generalization to larger populations
- Closed questions and logged interaction data produce quantitative data
- Be careful: mean, median, and mode can tell very different stories for the same dataset

Qualitative

- Difficult to measure sensibly as numbers — themes, narratives, patterns
- Captures the 'why' and 'how' behind behaviors
- Expressed as themes, patterns, stories, or critical incidents
- Open questions, interviews, and field notes produce qualitative data
- Analysis is interpretive; researcher judgment is integral

How Question Design Affects Analysis

- Open Questions — Each answer must be analyzed individually; rich but labor-intensive
- Closed Questions — Analyzed quantitatively; efficient but restrict what can be said
- Fixed alternative answers constrain findings — users cannot express nuances outside provided options
- Design tip: Use open questions in pilot studies to generate categories, then convert to closed questions for large-scale deployment
- The analysis you can perform is fundamentally limited by how you gathered the data

Basic Qualitative Analysis Strategies

- Critical Incidents — Focus on key events that reveal important behaviors or breakdowns
- Inductive Analysis (Thematic) — Themes emerge from the data without pre-existing framework
- Deductive Analysis (Categorical) — Data is sorted into pre-specified categories from theory or prior research
- In Practice — Most studies use a combination: start deductively, then allow inductive themes to emerge
- Granularity matters — Analysis can occur at word-level, phrase-level, utterance-level, or event-level

Analytical Frameworks: Matching Method to Goal

- Conversation Analysis — Recordings of spoken talk; focuses on how conversations are managed (pauses, turn-taking, inflection)
- Discourse Analysis — Speech or writing; examines how words convey meaning and construct reality; no objective 'truth' assumed
- Content Analysis — Any 'text' (video, audio, images); classifies into themes and studies frequencies
- Interaction Analysis — Video of natural activity; studies verbal and non-verbal interactions between people and artifacts
- Grounded Theory — Develops theory systematically from empirical data through open, axial, and selective coding
- Systems-Based Frameworks — Macro-level analysis of socio-technical systems (hospitals, airports, organizations)

Discourse, Content & Interaction Analysis

Discourse Analysis

- Focuses on dialogue and meaning-making
- Assumes language is a constructive tool, not a neutral conveyor
- Useful for identifying subtle, implicit, or hidden meanings
- Interpretive; researcher brings theoretical lens

Content & Interaction Analysis

- Content Analysis: classify data into themes/categories; count frequencies
- Can analyze any 'text': video, newspapers, ads, images, sounds
- Interaction Analysis: collaborative, inductive team process using video
- Researchers log contents, extract salient events, assemble instances, and study assemblages together

Grounded Theory: Building Theory from Data

- Goal — Develop theory systematically from empirical data rather than testing pre-existing hypotheses
- Open Coding — Identify initial categories and concepts from raw data
- Axial Coding — Flesh out categories and link them to subcategories; find relationships
- Selective Coding — Form a theoretical scheme around a central phenomenon
- Researchers draw on their theoretical backgrounds to inform analysis (not purely objective)
- Analytic tools: question the data, analyze words/phrases/sentences, make constant comparisons

Tools to Support Data Analysis

Quantitative Tools

• Spreadsheets (Excel, Google Sheets) — simple graphs and descriptive stats

• Statistical packages: SPSS, SAS, R, Python (pandas, scipy)

• Web analytics: Google Analytics, Hotjar, Mixpanel

• Interaction logging tools: custom instrumentation

Qualitative Tools

• NVivo — coding, categorization, theme-based analysis

• Dedoose — mixed-methods platform; web-based collaboration

• Atlas.ti — advanced qualitative coding and visualization

• CAQDAS — Computer Assisted Qualitative Data Analysis; University of Surrey networking project

• Many tools now support quantitative analysis of text-based qualitative data (sentiment, frequency)

Presenting Findings Effectively

- Structured Notations — Use clear syntax and semantics (e.g., affinity diagrams, journey maps, flowcharts)
- Stories and Narratives — Easy, intuitive way to communicate user experiences and design implications
- Summarize using a range of notations: statistical summaries, quotes, personas, scenarios, visualizations
- Quantitative data: use appropriate averages (mean vs. median vs. mode) and show variability
- Qualitative data: use representative quotes; ensure anonymity; provide sufficient context
- Always connect findings back to the original research questions and design implications

Common Pitfalls in Data Analysis & Interpretation

- Confusing correlation with causation — just because two variables move together does not mean one causes the other
- Overgeneralizing from small or biased samples — qualitative depth does not excuse poor sampling
- Ignoring outliers without investigation — outliers may reveal critical incidents or system failures
- Confirmation bias — seeking data that supports preconceptions while dismissing contradictory evidence
- Misusing averages — mean can be skewed by extreme values; median often better for skewed distributions
- Cherry-picking quotes — presenting atypical quotes as representative misleads stakeholders

PART 3

Data at Scale: Big Data in HCI

What is Data at Scale?

- Also known as Big Data — describes massive collections of numbers, images, conversations, videos, sensor streams
- Sources: social media posts, mobile sensor data, facial recognition, transaction logs, IoT devices, clickstreams
- Huge potential for solving complex problems: predicting disease outbreaks, optimizing traffic, personalizing education
- Critical danger: privacy erosion — data from multiple sources can be combined to reveal behavior patterns individuals never consented to share
- The Heathrow airport tracking example: security monitoring can violate traveler privacy

Collecting Data at Scale: Methods & Sources

- Web Scraping — Automated extraction of data from websites; raises legal and ethical questions
- Crowdsourcing — Amazon Mechanical Turk, citizen science; quality control is essential
- Personal Data Collection — Direct (surveys) and indirect (fitness trackers, smartphones, smart home devices)
- Sentiment Analysis — Automated classification of emotional tone in text (social media, reviews)
- Social Network Analysis — Mapping relationships and influence patterns (e.g., voting patterns, information diffusion)
- Sensor Networks — Environmental sensors, wearables, autonomous vehicles generate continuous data streams

Making Data at Scale Meaningful

- Raw big data is overwhelming — visualization and aggregation are essential for sensemaking
- Time-series analysis reveals trends: e.g., student stress levels spike before deadlines, sleep drops during exams
- Network graphs show relationships: clusters, bridges, and isolated nodes in social or information networks
- Heatmaps and treemaps surface density and hierarchy in large categorical datasets
- Dashboards combine multiple visualizations for real-time monitoring and decision support
- Key challenge: avoiding 'chart junk' — every visual element should serve a communicative purpose

Ethics in Data at Scale: The FATE Framework

- Fairness — Is the treatment just and without favoritism or discrimination? Are algorithms biased against certain groups?
- Accountability — Is the data accurate and correct? Who is responsible when automated decisions cause harm?
- Transparency — Are the decisions being made by a system visible to those affected by them?
- Explainability — Can people understand the explanations provided by the system? Is the reasoning accessible?
- Goal of FATE — Make decisions made by computer systems just, fair, and comprehensible to humans
- In HCI design, FATE is not an afterthought; it must be embedded from the earliest data collection stages

PART 4

From Data to Design: Requirements

What Are Requirements?

- A requirement is a statement about an intended product specifying what it should do or how it should perform
- Requirements exist at different levels of abstraction — from high-level goals to detailed technical specifications
- User Stories (Agile): 'As a <role>, I want <behavior> so that <benefit>'
- Example: 'As a traveler, I want to save my favorite airline so that I can collect air miles'
- Formal templates (e.g., Volere) capture: description, rationale, source, fit criterion, satisfaction, dissatisfaction, dependencies, conflicts
- Well-articulated requirements prevent miscommunication between designers, developers, and stakeholders

Types of Requirements

- Functional — What the system should do (features, capabilities, behaviors)
- Data — What kinds of data need to be stored and how (database schema, formats, retention)
- Environmental / Context of Use — Physical (dust, noise, light), Social (collaboration, privacy), Organizational (training, support), Technical (platforms, compatibility)
- User Characteristics — Nationality, education, attitudes, expertise level (novice vs. expert vs. casual vs. frequent)
- Usability Goals — Efficiency, effectiveness, safety, utility, learnability, memorability
- User Experience Goals — Enjoyment, fun, aesthetics, trustworthiness, motivation

Data Gathering for Requirements

Techniques

- Interviews — explore needs and frustrations directly
- Observation — see what users actually do vs. what they say
- Questionnaires — quantify prevalence of needs across populations
- Documentation Review — understand rules, procedures, legislation; no stakeholder time required
- Competitive Analysis — study similar products to prompt requirements

Combination Strategies

- Diaries + Interviews — temporal patterns + deep explanation
- Observation + Think-Aloud + Questionnaire — behavior + cognition + attitude
- Ethnography + Usability Tests + Participation — deep context + validated interaction
- Documentation + Evaluation + Observation + Group Interviews — comprehensive organizational view

Contextual Inquiry & Technology Probes

- Contextual Inquiry — One-on-one field interviews (1.5–2 hours) using a master/apprentice model
- Four Principles: Context (go to the user), Partnership (explore together), Interpretation (co-interpret), Focus (project-driven attention)
- Cool Concepts: Joy of Life (accomplish, connection, identity, sensation) and Joy of Use (direct action, hassle factor, learning delta)
- Technology Probes — Deploy toolkits, apps, or sensors to see how users appropriate technology in their own context
- Provocative Probes — Designed to challenge norms and attitudes (e.g., a device that makes laundry practices visible to provoke reflection)

Brainstorming for Innovation

- Include participants from a wide range of disciplines and experience levels
- Do not ban silly ideas — they often spark legitimate innovations through analogy or recombination
- Use catalysts for inspiration: analogous domains, extreme users, constraint removal
- Keep records of every idea without censoring during the session
- Sharpen the focus — a well-defined 'How Might We' question prevents aimless ideation
- Use warm-up exercises and make the session fun; psychological safety drives creativity

Personas: Bringing Users to Life

- Personas are rich descriptions of typical users — synthesized from real user research, not fictional guesses
- They capture a user profile: attitudes, motivations, goals, frustrations, skills, and context
- Personas should be typical, not idealized — include realistic limitations and pain points
- Bring to life with: name, photo, demographics, background story, goals, and quotable quotes
- Develop a small set (3–5) with one primary persona who represents the core user group
- A good persona helps the design team make decisions by asking: 'Would this work for [Name]?'

Scenarios: Narratives of Use

- Scenarios are informal narrative stories describing how a persona interacts with a product in a specific context
- Characteristics: simple, natural, personal, and not ger
- Structure: Persona (who) + Scenario context (when/w sequence
- Scenarios explore existing behavior, potential new pr
- They bridge the gap between abstract requirements a
- Design Fiction extends scenarios into speculative futu consequences

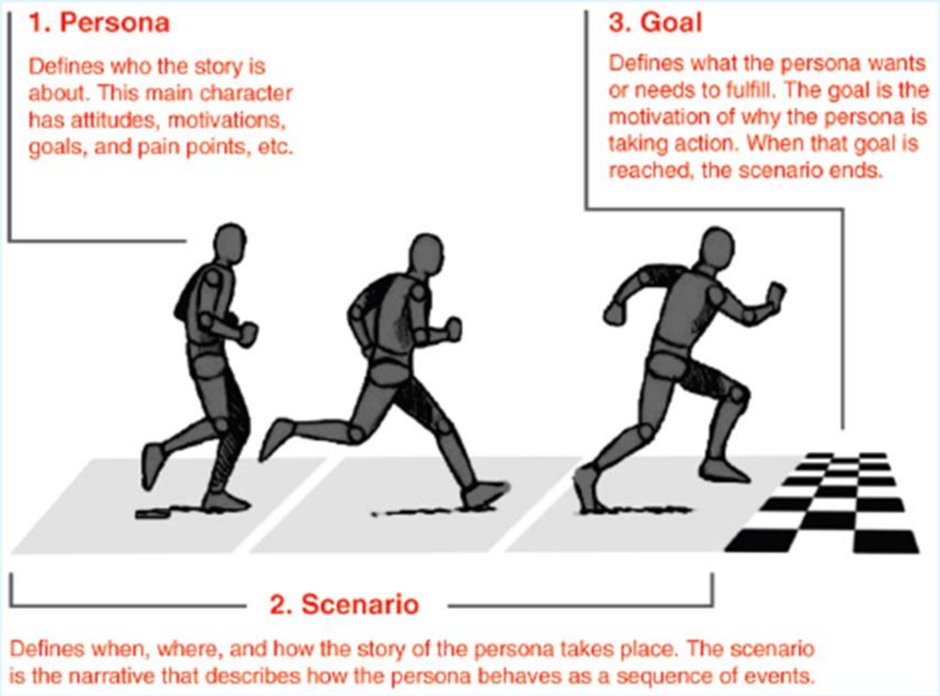

Figure 10.10 The relationship between a scenario and its associated persona
Source: http://www.smashingmagazine.com/2014/08/06/a-closer-look-at-personas-part-1/

Scenarios: Narratives of Use of the Personas with Goals

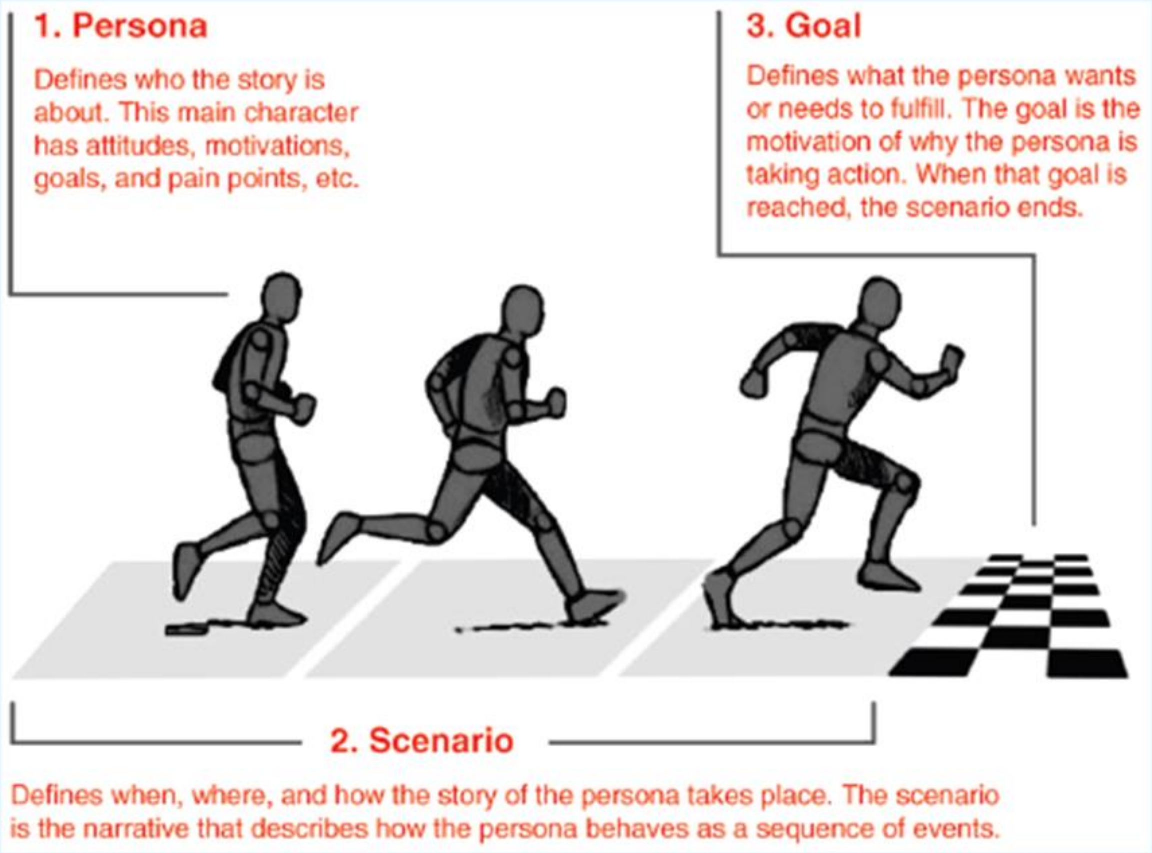

Figure 10.10 The relationship between a scenario and its associated persona

Source: http://www.smashingmagazine.com/2014/08/06/a-closer-look-at-personas-part-1/

Use Cases: Capturing Interaction

Essential Use Cases

- Abstract description of user intention and system responsibility
- No implementation detail — focuses on task division
- Example: USER INTENTION 'Find visa requirements' → SYSTEM RESPONSIBILITY 'Request destination and nationality'
- Useful early in design when technology choices are undecided

Detailed Use Cases

- Step-by-step numbered sequence of interactions
- Normal course: the 'happy path' through the interaction
- Alternative courses: error handling, edge cases, exceptions
- Example: If country name is invalid → error message → return to step 1
- More detail makes them suitable for developer handoff

Usable Security: A Critical Requirement

- Security mechanisms that are unusable will be circumvented by users — undermining the very security they provide
- Passwords are the canonical example: too much advice leads to coping strategies (writing down, reusing) that weaken security
- Design principle: make the secure path the easy path — reduce friction for compliant behavior
- Consider: biometric authentication, single sign-on, password managers, and risk-based adaptive authentication
- Usability and security are not opposed; good HCI design achieves both simultaneously

From Data to Design: The Synthesis Pipeline

- Step 1: Gather — Interviews, observation, questionnaires, probes, contextual inquiry
- Step 2: Analyze — Code themes, find patterns, identify critical incidents, compute metrics
- Step 3: Characterize — Build personas that embody the user types discovered in analysis
- Step 4: Narrate — Write scenarios that place personas in realistic contexts of use
- Step 5: Specify — Derive functional, data, environmental, and usability requirements from scenarios
- Step 6: Validate — Test requirements with stakeholders; iterate when gaps are found

Data Gathering and Sensemaking: A Synthesis

- Data Gathering (Ch. 8) provides the raw material — but material without analysis is just noise
- Analysis (Ch. 9) transforms raw data into structured insights — themes, patterns, and evidence
- Data at Scale (Ch. 10) extends these methods to massive, heterogeneous datasets with new ethical imperatives
- Requirements (Ch. 11) translate insights into designable, testable statements about the intended product
- Personas and scenarios are the bridge — they keep human context alive through technical specification
- The ultimate goal is not data, but understanding; not understanding, but better human-computer interaction

Key Takeaways for CSE 4849

- Plan before you gather: goals, participants, methods, and analysis strategy must align
- Triangulate methods and data sources — no single technique tells the whole story
- Qualitative and quantitative data are complementary, not competing; combine them wisely
- Ethics (FATE) must be designed into data collection, not retrofitted after the fact
- Requirements are living documents — they evolve as understanding deepens through iteration
- The best HCI researchers are sensemakers: they turn messy human behavior into clear, actionable design

References & Further Reading

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